

What Changed 10 October 2026

New features and changes, and how to test each one

October 2026 · For the Polynexus team

THE JOURNEY AT A GLANCE

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Before you start

Each part says what changed, then **how to test it**. Use the local console (<http://localhost:3000> , backend on port 8000) for SaaS features, and the trial install ([C:\hms-test](#) , see TRIAL_INSTALL_WINDOWS.pdf) for on-premise features.

Before testing licence features, turn on 2FA for your SaaS Owner account: Settings → Two-factor authentication. Every licence step asks for a code.

1 Permissions

What changed

- Every hospital role has a defined set of permissions (the permission matrix); every screen and API checks it.
- Rules for who may assign which role: an administrator can't give someone more rights than they have.
- SaaS console → a hospital → *permissions*: Polynexus can cap what a hospital's roles may ever be given (the "ceiling"), inside the modules it bought.

How to test

1. In a hospital, create a user with the *Receptionist* role and sign in as them: clinical and finance menus are hidden, and opening a hidden page's address directly shows *no access*.
2. As a hospital administrator, try to give a user a role with more rights than your own: it's refused.
3. In the SaaS console, set a hospital's permission ceiling to exclude a permission, then check a role in that hospital can't use it.

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Licensing

What changed

- A licence lists **features** (HMS core, CRM, ERP, MIS, WhatsApp, IVR, AI assist, NABH, multi-branch, patient portal, API access). Unlicensed modules vanish from menus and their APIs refuse requests.
- Each installation has a **deployment ID**; a licence works only on its own deployment.
- Optional **hardware binding** to one machine's fingerprint.
- Expiry: yellow banner 30 days before, red during the grace period, then **read-only** (view and print only; data never deleted).
- Checked at start-up, on every request (cached 10 minutes) and hourly.

How to test (trial install)

1. Issue a licence with HMS core, ERP and AI assist only. CRM menus (Enquiries, Call console, Callbacks) don't appear.
2. Settings → License shows the features, user limit, expiry, deployment ID and *Bound to this server*.
3. Issue a licence for a different deployment ID and upload it: it's refused with *This license is for deployment*
4. Edit one character of a `.lic` file and upload it: *The license signature is not valid*.

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Licence security

What changed — the nine safeguards:

#	Safeguard
1	Blocking a user ends all their sessions immediately (not at next login)
2	Only SaaS Owners sign licences. Platform Admins lost licence issuing; other staff can only request, and only if an Owner allows it
3	A fresh 2FA code is needed to request, approve, grant the right and block
4	Two-person rule: a request waits until a SaaS Owner approves it
5	Every staff account has a permanent code (<code>PNX-0001</code> ...). The requester's and approver's codes are signed into each licence and shown in the register, in <code>verify</code> , and in the hospital's Settings → License
6	Owners are emailed for every request and every licence; the register flags <i>No paid invoice</i>
7	The private key is only on the SaaS server; the command-line tool can only verify
8	A signed revocation list is built into each release: revoked licences stop on upgrade
9	A usage report naming a licence the console never issued is refused and raises a critical security alert

Also: Owner emails now go out over SMTP. Set `EMAIL_HOST` , `EMAIL_PORT` , `EMAIL_HOST_USER` , `EMAIL_HOST_PASSWORD` , `EMAIL_USE_TLS` and `DEFAULT_FROM_EMAIL` on the SaaS server.

How to test (local console; you need a second staff account, e.g. a Platform Admin, with 2FA on)

- 1. Platform Admin can't issue.** Sign in as the Platform Admin → a hospital → License → fill in, 2FA code → *Request: Only SaaS Owners, and staff a SaaS Owner has authorised, can issue licences.*
- 2. Owner grants the right.** As Owner: On-Premise Licences → Licence team → the admin's row → *Allow licences* → 2FA. Row shows *May request*.
- 3. Request waits.** As the admin, request again: *Sent to a SaaS Owner for approval*. No licence appears in the register. The Owner gets an email (in local development it's printed in the backend window).
- 4. Wrong code refused.** Type `000000` as the 2FA code: *Enter the current 6-digit code from your authenticator app*.
- 5. Owner approves.** As Owner: Licence requests → *Approve* → 2FA. The file downloads. The register shows *Issued by* = the admin's code, *approved* = yours, and *No paid invoice* (unless the hospital has a paid invoice).
- 6. Codes inside the file.** `python -m license_issuer verify <file>` prints `Issued by : PNX-... (approved by PNX-...)` . Install it on the trial server: Settings → License shows *Issued by*.
- 7. CLI can't issue.** `python -m license_issuer new` prints that issuing is done in the SaaS console, and stops.
- 8. Blocking ends sessions.** Keep the admin signed in in another browser. As Owner: Licence team → *Block* → reason → 2FA. The admin's next click signs them out. Their right is removed; their pending requests show *cancelled*.
- 9. Revocation.** Revoke a licence in the register. Click *Revocation list*, save it as `Backend/apps/licensing/revocations.lic` , build a bundle and upgrade the trial install: Settings → License shows *Revoked* and the system is read-only.
- 10. Unknown licence alert.** Upload a usage report from an installation whose licence isn't in the register (or change the licence ID inside a report): the upload is refused with *ALERT ...*, Owners get an email, and a critical security event is logged.

What changed

- **Settings** → **Subscription** (hospital side, SaaS mode): plan, billing cycle, modules, next billing date, invoices with PDF download.
- **On-Premise Licences** tab: every licence, renewals due first; filters (active, expired, revoked, expiring in 60 days); search by hospital, licence ID or staff code; who issued and approved; paid flag; last usage report.
- **Usage reports**: the hospital downloads one in Settings → License (counts only, no patient details, sealed against editing); you upload it in On-Premise Licences.
- **Licence requests** and **Licence team** panels (Part 3).

How to test

1. Sign in as a hospital administrator (SaaS mode): Settings → Subscription shows the plan and invoices; an invoice PDF downloads.
2. On the trial install: Settings → License → *Download usage report*. Upload it in the console: *Usage report added: N active users ...*, and the licence row shows the numbers.
3. Change a number inside the report file and upload it: *edited after the product created it*.
4. Filter *Expiring in 60 days* and search by a staff code.

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Install bundle

What changed

- Production images: backend compiled with Cython (no source code), frontend minified and obfuscated, nginx as web server and proxy.
- `make bundle` produces one `bundle-x.y.z.tar.gz` with everything an offline server needs, including PostgreSQL, Redis and the 3D model (Git LFS). The build stops if the real model wasn't pulled.
- Installers for Linux (`install.sh`) and Windows (`install.ps1`): offline image load, fresh secrets, deployment ID, fingerprint, `REQUEST-LICENCE.txt` , licence activation, administrator creation. They pick another port if 80/443 are taken.
- `upgrade.sh` / `upgrade.ps1` : back up the database first, then update and restart, keeping `.env` , licence and data.
- `make bundle` warns if there's no revocation list.

How to test

1. Follow TRIAL_INSTALL_WINDOWS.pdf end to end.
2. Build a second bundle with a higher version and run `upgrade.ps1 -InstallDir C:\hms-test : a .dump` backup appears in `backups/` , the version changes, data stays.
3. Inside the backend container, check there are no `.py` source files for the apps: `docker compose exec web sh -c "find /app/apps -name '*.py' | grep -v __init__ | head"` should show (almost) nothing.

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Domain and backups

What changed

- `configure-domain.sh` / `.ps1` : set a domain such as `hms.hospital.com` with the hospital's own certificate, Let's Encrypt, or plain HTTP for internal use.
- Nightly backup at 02:00: database and documents, 30 days kept, status in `backups/LAST-BACKUP.txt` , `BACKUP_DIR` can point to a NAS or external drive.

How to test

1. On the trial install: `docker compose exec backup sh /backup.sh now` , then open `backups/LAST-BACKUP.txt` : it says OK.
2. Run `configure-domain.ps1` with a test name (for example `hms.test.local` added to the Windows hosts file) and option 3: the site opens at that name.

What changed

- **Patient audit trail:** creating, changing or deleting a patient records which fields changed, by whom, when and from where. Encrypted details (identifiers, phone, address) are recorded as *changed* without their values, so the log never holds them in clear. View it in Admin & audit → Audit Trail.
- **3D anatomy** moved to the HMS clinical menu. If the 3D model isn't installed, the page explains that instead of showing a blank screen.

How to test

1. Edit a patient's name and phone number, then open Admin & audit → Audit Trail: the entry shows your name, the time, the old and new name, and the phone number as *changed* (no number shown).
2. Open Anatomy from the HMS clinical menu: the 3D model loads.

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Documents

Document	Status
docs/HOW_IT_WORKS.pdf	New. The complete procedure for SaaS and on-premise
docs/CHANGES_2026-10-10.pdf	New. This document
docs/CLIENT_DELIVERY.pdf	Updated. Licences now via the console with approval; revocation list before building; staff leaving
docs/TRIAL_INSTALL_WINDOWS.pdf	Updated. Part 4 issues the licence in the local SaaS console
deploy/RUNBOOK.md	Updated. Licence team, approval, revocation, offboarding, new troubleshooting line
docs/COMPLIANCE_REVIEW.md	Written today: audit, encryption, DPDP, NABH, with open gaps

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Test checklist

#	Test	Part	OK
1	Receptionist can't see clinical/finance menus	1	[]
2	Unlicensed CRM menus hidden on trial install	2	[]
3	Edited licence file refused	2	[]
4	Platform Admin can't request without the right	3	[]
5	Request waits for Owner; Owner email received	3	[]
6	Wrong 2FA code refused	3	[]
7	Approved licence shows both staff codes (register, verify, Settings)	3	[]
8	<code>license_issuer</code> new refuses	3	[]
9	Blocking signs the person out at once	3	[]
10	Revoked licence stops after upgrade	3	[]
11	Unknown-licence usage report raises an alert	3	[]
12	Subscription card and invoice PDF	4	[]
13	Usage report upload; edited report flagged	4	[]
14	Trial install end to end	5	[]
15	Upgrade keeps data and makes a backup	5	[]
16	Manual backup says OK	6	[]
17	Patient audit trail shows a change	7	[]
18	Anatomy opens from the HMS menu	7	[]

Automated tests: backend `pytest` (licence controls in `apps/saas_admin/test_licence_controls.py`), frontend `npx vitest run`. Three older backend tests are known to fail and are unrelated (hospital switching for SaaS admins, one telephony adapter test).